

11.24

(a)

```
Call:
lm(formula = lprice ~ mon + tue + wed + thu + stormy + mixed,
    data = fultonfish)
```

```
Residuals:
    Min       1Q   Median       3Q      Max
-0.92281 -0.18819  0.00813  0.22404  0.68381
```

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.35957    0.07915  -4.543 1.50e-05 ***
mon          -0.10825    0.10339  -1.047  0.29753
tue          -0.06609    0.10103  -0.654  0.51446
wed          -0.04927    0.10377  -0.475  0.63591
thu           0.03935    0.10071   0.391  0.69681
stormy       0.44634    0.07921   5.635 1.51e-07 ***
mixed       0.23688    0.07851   3.017  0.00321 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3413 on 104 degrees of freedom
Multiple R-squared:  0.245,    Adjusted R-squared:  0.2014
F-statistic: 5.624 on 6 and 104 DF, p-value: 4.349e-05
```

$p\text{-value} = 0.0032 < 0.05$, mixed is statistically significant at 5% level

$$H_0: \pi_5 = \pi_6 = 0$$

$$F = 16.14 > 10, p\text{-value} < 0.05$$

We reject H_0 . Stormy and mixed are jointly significant.

```
Linear hypothesis test:
stormy = 0
mixed = 0
```

```
Model 1: restricted model
Model 2: lprice ~ mon + tue + wed + thu + stormy + mixed
```

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	106	15.876				
2	104	12.115	2	3.7604	16.14	7.857e-07 ***

(b)

```
Call:
ivreg(formula = lquan ~ lprice + mon + tue + wed + thu | mon +
       tue + wed + thu + stormy + mixed, data = fultonfish)
```

```
Residuals:
    Min       1Q   Median       3Q      Max
-2.1327 -0.4320  0.0654  0.5133  1.1798
```

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  8.54023    0.15661  54.533 < 2e-16 ***
lprice       -0.93014    0.35340  -2.632  0.00977 **
mon          -0.01190    0.20837  -0.057  0.95456
tue          -0.52583    0.20230  -2.599  0.01068 *
wed          -0.56262    0.20696  -2.719  0.00767 **
thu           0.09987    0.20285   0.492  0.62350
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6853 on 105 degrees of freedom
Multiple R-Squared:  0.185,    Adjusted R-squared:  0.1462
Wald test: 4.927 on 5 and 105 DF, p-value: 0.0004317
```

$$\hat{\alpha}_1 = -0.9301$$

Table 11.5, $\hat{\alpha}_1 = -1.1194$.

After adding mixed, the estimated demand elasticity becomes smaller.

(c)

```
Call:
ivreg(formula = lquan ~ lprice + mon + tue + wed + thu | mon +
       tue + wed + thu + stormy + mixed, data = fultonfish)
```

```
Residuals:
    Min       1Q   Median       3Q      Max
-2.1327 -0.4320  0.0654  0.5133  1.1798
```

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  8.54023    0.15661  54.533 < 2e-16 ***
lprice       -0.93014    0.35340  -2.632  0.00977 **
mon          -0.01190    0.20837  -0.057  0.95456
tue          -0.52583    0.20230  -2.599  0.01068 *
wed          -0.56262    0.20696  -2.719  0.00767 **
thu           0.09987    0.20285   0.492  0.62350
```

```
Diagnostic tests:
            df1 df2 statistic p-value
Weak instruments  2 104    16.140 7.86e-07 ***
Wu-Hausman      1 104     1.489   0.225
Sargan          1  NA     0.772   0.380
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6853 on 105 degrees of freedom
Multiple R-Squared:  0.185,    Adjusted R-squared:  0.1462
Wald test: 4.927 on 5 and 105 DF, p-value: 0.0004317
```

$$J = 0.7722, p\text{-value} = 0.380 > 0.05$$

We do not reject H_0 . It is no evidence to conclude that the surplus instrument is invalid.

(d)

Linear hypothesis test:

```
mon = 0
tue = 0
wed = 0
thu = 0
```

```
Model 1: restricted model
Model 2: lprice ~ mon + tue + wed + thu + stormy + mixed
```

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	108	12.408				
2	104	12.115	4	0.29256	0.6279	0.6437

$$H_0: \pi_1 = \pi_2 = \pi_3 = \pi_4 = 0$$

$$F = 0.6279, p = 0.6437 > 0.05$$

We do not reject H_0 . It means MON, TUE, WED and THU are not jointly significant. This suggests that the weekday dummy variables are weak instruments.

11.28

$$(a) D: Q = \alpha_1 + \alpha_2 P + \alpha_3 PS + \alpha_4 DI + e_d$$

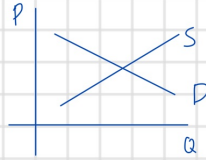
$$S: Q = \beta_1 + \beta_2 P + \beta_3 PF + e_s$$

$$\Rightarrow D: P = \delta_1 + \delta_2 Q + \delta_3 PS + \delta_4 DI + u_d$$

$$S: P = \gamma_1 + \gamma_2 Q + \gamma_3 PF + u_s$$

For demand, $\delta_2 < 0$, $\delta_3 > 0$, $\delta_4 > 0$

For supply, $\gamma_2 > 0$, $\gamma_3 > 0$



(b) Call: `ivreg(formula = p ~ q + ps + di | pf + ps + di, data = truffles)`

Residuals:

	Min	1Q	Median	3Q	Max
	-39.661	-6.781	2.410	8.320	20.251

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-11.428	13.592	-0.841	0.40810
q	-2.671	1.175	-2.273	0.03154 *
ps	3.461	1.116	3.103	0.00458 **
di	13.390	2.747	4.875	4.68e-05 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 13.17 on 26 degrees of freedom
Multiple R-squared: 0.5567, Adjusted R-squared: 0.5056
Wald test: 17.37 on 3 and 26 DF, p-value: 2.137e-06

Call: `ivreg(formula = p ~ q + pf | ps + di + pf, data = truffles)`

Residuals:

	Min	1Q	Median	3Q	Max
	-9.7983	-2.3440	-0.6281	2.4350	11.1600

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-58.7982	5.8592	-10.04	1.32e-10 ***
q	2.9367	0.2158	13.61	1.32e-13 ***
pf	2.9585	0.1560	18.97	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.399 on 27 degrees of freedom
Multiple R-squared: 0.9486, Adjusted R-squared: 0.9448
Wald test: 232.7 on 2 and 27 DF, p-value: < 2.2e-16

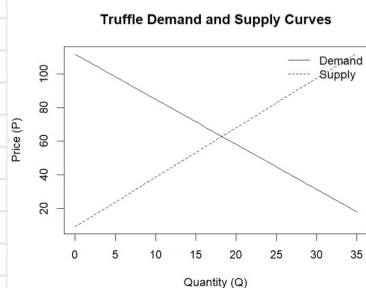
$$D: \hat{p} = -11.4284 - 2.671 Q + 3.461 PS + 13.390 DI$$

$$S: \hat{p} = -58.7982 + 2.9367 Q + 2.9585 PF$$

Yes, the signs are correct. All coefficients are significantly different from zero.

$$(c) \text{ Demand elasticity} = -1.2725$$

(d)



Demand: negative slope

Supply: positive slope.

Demand and supply curves have the expected slopes.

$$(e) Q_{EQM} = 18.2503, P_{EQM} = 62.8427$$

$$Q_{EQM,RF} = 18.2604, P_{EQM,RF} = 62.8154$$

Structural and reduced-form equilibrium values agree closely.

```
> q_eq
(Intercept)
18.25021
> p_eq
(Intercept)
62.84257
```

```
> q_eq_rf
1
18.2604
> p_eq_rf
1
62.81537
```

(f)

Call: `lm(formula = p ~ q + ps + di, data = truffles)`

Residuals:

	Min	1Q	Median	3Q	Max
	-25.0753	-2.7742	-0.4097	4.7079	17.4979

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-13.6195	9.0872	-1.499	0.1460
q	0.1512	0.4988	0.303	0.7642
ps	1.3607	0.5940	2.291	0.0303 *
di	12.3582	1.8254	6.770	3.48e-07 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 8.814 on 26 degrees of freedom
Multiple R-squared: 0.8013, Adjusted R-squared: 0.7784
F-statistic: 34.95 on 3 and 26 DF, p-value: 2.842e-09

Call: `lm(formula = p ~ q + pf, data = truffles)`

Residuals:

	Min	1Q	Median	3Q	Max
	-8.4721	-3.3287	0.1861	2.0785	10.7513

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-52.8763	5.0238	-10.53	4.68e-11 ***
q	2.6613	0.1712	15.54	5.42e-15 ***
pf	2.9217	0.1482	19.71	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.202 on 27 degrees of freedom
Multiple R-squared: 0.9531, Adjusted R-squared: 0.9496
F-statistic: 274.4 on 2 and 27 DF, p-value: < 2.2e-16

$$D: \hat{p} = -13.6195 + 0.1512 Q + 1.3607 PS + 12.3582 DI \Rightarrow \text{coefficient on } Q \text{ is positive, OLS demand equation has the wrong sign.}$$

$$S: \hat{p} = -52.8763 + 2.6613 Q + 2.9217 PF \Rightarrow \text{signs are correct, the coefficients are statistically significant.}$$

$\therefore$ 2SLS is more appropriate for solving the endogeneity problem.